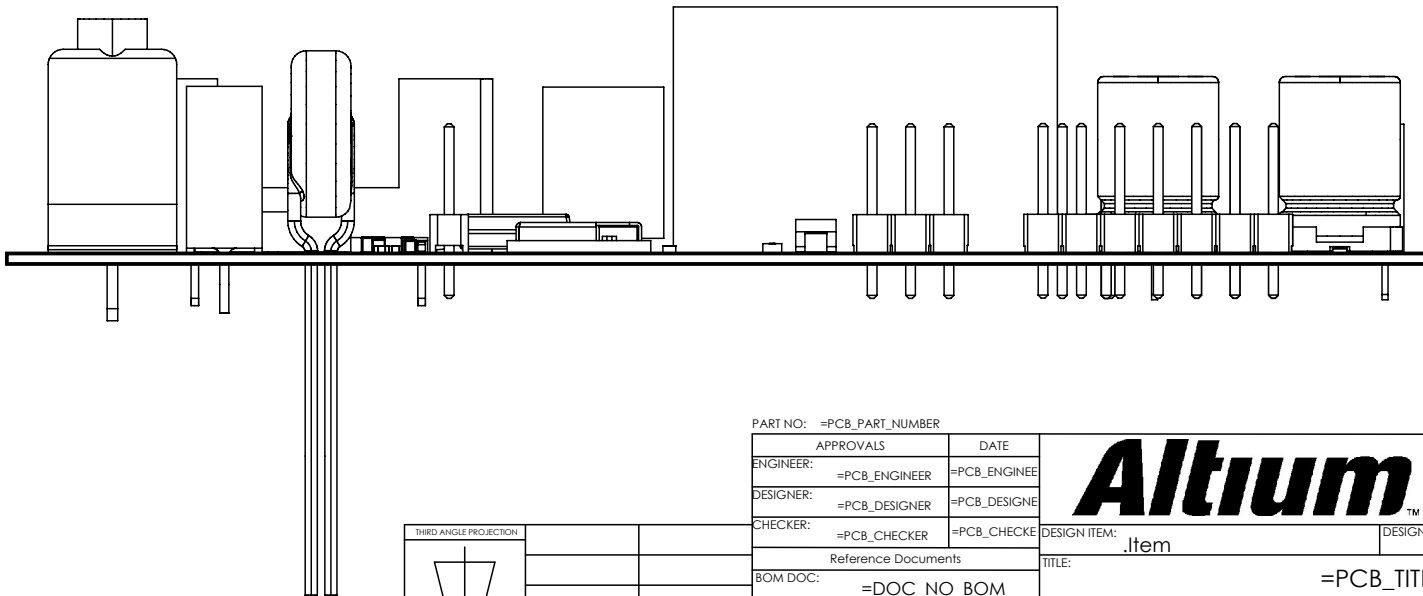
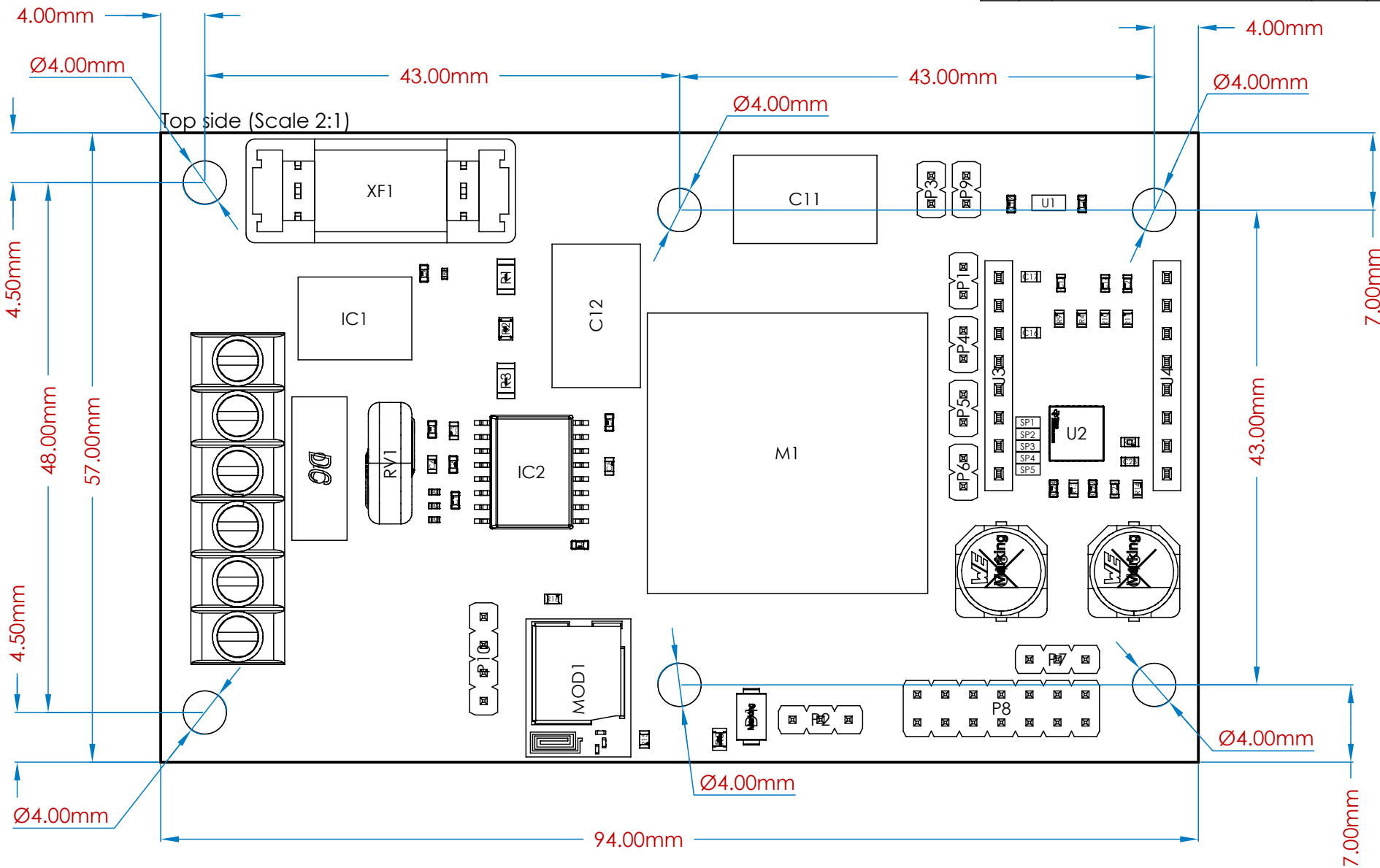
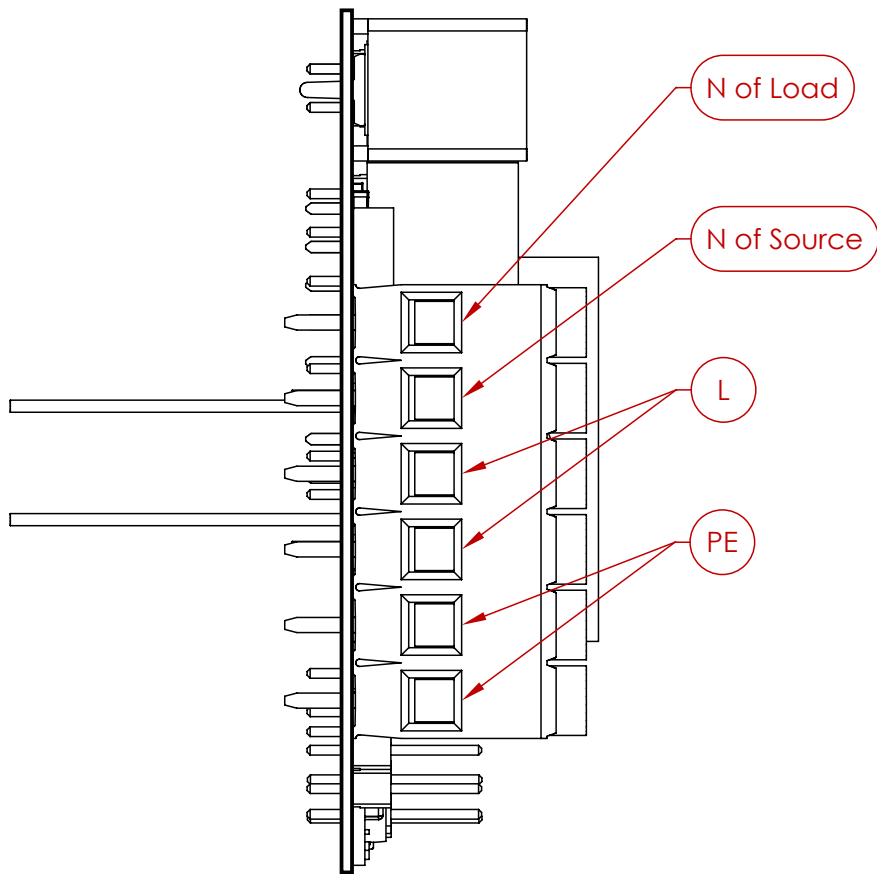


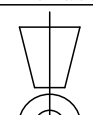
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REV STATUS OF SHEETS	REV	-	-	-	-	-	-	-	-	-	REVISIONS				
	SHEET	-	-	-	-	-	-	-	-	-	ZONE	REV	DESCRIPTION	DATE	APPROVED
											--	-	-----	--/--/--	-, -----
											--	-	-----	--/--/--	-, -----
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NOTES UNLESS OTHERWISE SPCEIFIED:

1. Board shape optimized for BOPLA SE 435 DE - 9016 housing.
2. NOTE
3. NOTE

		PART NO: =PCB_PART_NUMBER		Altium™			
		APPROVALS	DATE				
		ENGINEER: =PCB_ENGINEER					
		DESIGNER: =PCB_DESIGNER					
		CHECKER: =PCB_CHECKER					
THIRD ANGLE PROJECTION					DESIGN ITEM: .Item	DESIGN ITEM REVISION: .ItemRevision	
		Reference Documents		TITLE:			
		BOM DOC: =DOC_NO_BOM		=PCB_TITLE_1			
		ASSY DOC: =DOC_NO_FAB_DWG		=DRAF =PCB_TITLE_2			
		SCH DOC: =DOC_NO_SCH_DWG		SIZE: A3	CAGE CODE: =CAGE_CO	DWG NO: =DOC_NO_ASSY_DWG	REV:
		PCB DOC: =PCB_DWG_NO		SCALE: New	FILE NAME: AC-Power-Monitor.PCBdwf		SHEET: 1 OF 5
NEXT ASSY		USED ON					
APPLICATION							

Bill Of Materials

Line #	Designator	Name	Quantity	Purpose
	C1	CC0603KRX7R9BB103	1	
	C2, C9, C19, C20, C22	CGA3E1X7R1E105K080AC	5	
	C3, C6, C8	06035C102KAT2A	3	
	C4, C5, C7, C24	CGA3E2X7R1H104K080AA	4	
	C10	10nF	1	
	C11, C12	MKY22W21003I00	2	
	C13, C14, C21	06035C103KAT4A	3	
	C15	180µF	1	
	C16, C17	C0603C105K3RACTU	2	
	C18	180µF	1	
	C23	885012206085	1	
	C25	C0603C474K4RACTU	1	
	D1	400W, 54VDC	1	
	FB1, FB2, FB3, FB4	1800Ohm	4	
	IC1	TMCS1127C1AQDVGR	1	
	IC2	AMC3330DWER	1	
	J1	691131710006	1	
	J3, J4	61300811821	2	
	M1	AC/DC-Buck-Module	1	
	MOD1	WIRL-COMB_2617011025000_1	1	
	P1, P3, P4, P5, P6, P9	61300211121	6	3.3V to ESP32, Bridge 5.0V to 3.3V, A1N to ESP32, A1P to ESP32, A0 to ESP32, 5.0V to ESP32
	P2, P7	61300311121	2	Power-Outputs, JTAG- Power Selector
	P8	61301421121	1	JTAG-Interface
	P10	61300411121	1	WLAN-BOOT- Interfaxce
	R1, R3	RK73H2BTTD1804F	2	
	R2	CRCW08054K70FKEA	1	
	R4	CRCW060347K0FKEA	1	
	R5	CRGCQ0805F1M0	1	
	R9	ERA-3AEB1472V	1	
	R10, R11	RN73C1J10KBTG	2	
	R12	KTR03EZPF4702	1	
	R13	CPF0603B20KE1	1	
	R14	AC0603FR-071KL	1	
	R15, R16	CRCW06031K50FKEA	2	
	RV1	275VAC, 350VDC	1	
	U1	MCP1700T-3302E/TT	1	
	U2	MSP430I2041TRHBR	1	
	XF1	696106003002	1	

Layer Stack Legend

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.01mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.04mm		Signal	GTL
Prepreg		0.07mm	PP-006	Dielectric	
CF-004	N / GND	0.07mm		Signal	G1
		0.32mm	FR-4	Dielectric	
CF-004	L / VCC	0.07mm		Signal	G2
Prepreg		0.07mm	PP-006	Dielectric	
Copper	Bottom Layer	0.04mm		Signal	GBL
Surface Material	Bottom Solder	0.01mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 0.69mm

Bill of Materials

Project Title: Bill of Materials for BCM Document [AC-Power-Monitor BomDoc]
Project File Name: AC-Power-Monitor PjPb
Assembly Variant: None

Altium
2175 Salk Avenue
Suite 200
Carlsbad, CA 92008
USA

Part Number	Description	Designator	Manufacturer	Manufacturer Part Number	Supplier	Supplier Part Number	Supplier Unit Price	Quantity
Custom Item 2	SE 435 DE - 9036		BOPLA	SE 435 DE - 9036	BOPLA	SE 435 DE - 9036	37.47	1
CMP-2100-03600	CAP CER 10000PF 50V X7R 0603	C1	Taiyo	C20630R1X7R9B8103	Mouser	823-C20630R1X7R9B8103	0.09	1
CMP-06246-0015	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C2, C22	TDK	CGA3E1X7R1E105K080AC	Mouser	810-CGA3E1X7R1E105K	0.14	2
CMP-2005-04378	General Purpose Ceramic Capacitor, 0603, 1uF, 10%, X7R, 15%, 50V	C3, C6	Kyocera AVX	8603SC102KAT2A	Mouser	981-0603SC102K	0.02	2
CMP-2000-07546	Cap Ceramic 1000F 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C4, C7	TDK	CGA3E2X7R1H104K080AA	Mouser	810-CGA3E2X7R1H104K	0.09	2
CMP-2000-07546	Cap Ceramic 1000F 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C5	TDK	CGA3E2X7R1H104K080AA	Mouser	810-CGA3E2X7R1H104K	0.09	1
CMP-2005-04378	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C8	Kyocera AVX	8603SC102KAT3A	Mouser	981-0603SC102K	0.02	1
CMP-06246-0015	WCAP-F17C2 Film Capacitor, THT, 1.13 x .05 x .111mm, 10uF, 275V	C9, C19, C20	TDK	CGA3E1X7R1E105K080AC	Mouser	810-CGA3E1X7R1E105K	0.14	1
8903240230063	Cap Ceramic 1uF 25V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C10, C12	Wurth	8903240230063	Wurth	8903240230063	0.38	1
C, Unipolar	CAP CER 10000PF 50V X7R 0603	C11, C12	WIMA	MM72W2100303K10D	Mouser	925-MM72W2100303K05	0.73	2
CMP-2005-02076	CAP CER 10000PF 50V X7R 0603	C13, C14, C21	Kyocera AVX	8603SC102KAT4A	Mouser	981-0603SC102KAT4A	0.01	3
87507515001, 87507515001, 87507515001	WCAP-P5LC Aluminum Polymer Capacitor, V-Chip SMT, D8 x H1.7mm, 180uF, 6.3V, WCAP-P5LC Aluminum Polymer Capacitor, V-Chip SMT, D8 x H1.7mm, 180uF, 10V	C15, C18	Wurth Electronics	87507515001, 87507515001	HePuram Digiplex	HePuram 732-6488-1-ND		2
CMP-2006-03713	87507515001, 87507515001, 87507515001	C16, C17	KEMET	87507515001, 87507515001	Mouser	89-03003C1063K	0.22	2
CMP-03913-5391	87507515001, 87507515001, 87507515001	C23	Wurth Electronics	87507515001, 87507515001	Digiplex	732-6003-1-ND	0.09	1
CMP-2000-07546	Cap Ceramic 1000F 50V X7R 10% Pad SMD 0603 +125°C Automotive T/R	C24	TDK	CGA3E2X7R1H104K080AA	Mouser	810-CGA3E2X7R1H104K	0.09	1
CMP-2006-03235	CAP CER 0.47UF 16V X7R 0603	C25	KEMET	C0603E474K46ACTU	Mouser	89-C0603E474K46	0.11	1
82450241	WE-TVSP SMT Power TVS Diode, DO-214AC, Bidirectional, 400W, 56VDC	D1	Wurth Electronics	82450241	Mouser	732-629201041	0.26	1
7429284182	WE-TMS2 Tiny Multilayer Suppression Bead, size 0402, 1800ohm, 0.21A	F81, F82, F83, F84	Wurth Electronics	7429284182	Digiplex	732-7408284182CT-ND	0.11	4
7429284182	WE-TMS2 Tiny Multilayer Suppression Bead, size 0402, 1800ohm, 0.21A	F81, F82, F83, F84	Wurth Electronics	7429284182	Digiplex	732-7408284182CT-ND	0.11	4
AMC3330DWER	Integrated Circuit	IC2	Texas Instruments	AMC3330DWER	Mouser	595-AMC3330DWER	5.70	1
6, 91132E-11	WR-TBL Series 1317 Horizontal Entry, THT, pitch 2.54mm, 1 Row, 4P	J1	Wurth Electronics	6, 91132E-11	Mouser	710-69132E13170006	0.69	1
6130011121	WR-PHD Socket Header, THT, Vertical, pitch 2.54mm, 1 Row, 4P	J3, J6	Wurth Electronics	6130011121	Mouser	710-6130011121	0.17	2
ACDC-Back-Module	Stephano-I - Combined 2.4 GHz Wi-Fi and Bluetooth® LE 5.0 Module	MD1	Texas Power	TMPS03-105	Mouser	895-TMPS03-105	55.33	1
6130011121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 2P	P1, P3, P4, P5, P6, P9	Wurth Electronics	6130011121	Mouser	710-6130011121	0.11	4
6130011121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 3P	P2, P7	Wurth Electronics	6130011121	Mouser	710-6130011121	0.11	2
61301421121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 2 Row, 14P	P8	Wurth Electronics	61301421121	Mouser	710-61301421121	0.85	1
61300411121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 1 Row, 4P	P10	Wurth Electronics	61300411121	Mouser	710-61300411121	0.17	1
CMP-2003-01283	1.8M Ω $\pm$ 1% 0.25W Chip Resistor 1206 Thick Film AEC-Q200 Qualified	R1, R3	VDA Speer	RK73HB87T1804F	Mouser	860-RK73HB87T1804F	0.09	2
CMP-2100-03615	RES Thick Film, 4.7k Ω , 1%, 1/10W 0603	R2	Vishay	C1CW085K470FKEA	Newark	53K0215	0.02	1
CMP-2000-03379	RES SMD 47k Ohm 1% 1/10W 0603	R4	Vishay	C1CW083K470FKEA	Mouser	71-C1CW083-47K-E3	0.09	1
CMP-03211-0324	CRCCQ0805 1M Ω 0.125W $\pm$ 100ppm/ $^{\circ}$ C 1% 125V	R5	TE Connectivity	CRCCQ0805F1M0	Arrow Electronics	270CQ0805F1M0	0.00	1
CMP-2000-00913	RES SMD 14.7k Ohm 0.1% 1/10W 0603	R6	Panasonic	ERA-3AB1472V	Digiplex	P14-70CBCT-ND	0.09	1
CMP-03211-0320	Thin Film Resistors RN 0603 50k Ω 0.1% 100ppm/ $^{\circ}$ C Res Thick Film 0603 47k Ω 1% 0.1W $\pm$ 100ppm/ $^{\circ}$ C Automotive Paper T/R	R10, R11	TE Connectivity	RN73C1108B1G	Digiplex	AJ55758-ND	0.09	2
CMP-08639-0308	RES SMD 20k Ω 0.1% 1/10W 0603	R12	Rohm	KTR082ZF4702	Mouser	755-KTR082ZF4702	0.14	1
CMP-03211-0066	RES SMD 20k Ω 0.1% 1/10W 0603	R13	TE Connectivity	OP6603030K1	Arrow Electronics	OP6603030K1	0.10	1
CMP-2000-04828	RES Thick Film, 1.3k Ω , 1%, 1/10W 0603	R14	Taiyo	ACD63PR-071N	Digiplex	211-141DCT-ND	0.09	1
CMP-2000-02095	RES Thick Film, 1.3k Ω , 1%, 1/10W 0603	R15, R16	Vishay	C1CW083K130FKEA	Annet	613M531	0.01	2
820412711	WE-VD Disk Varistor, THT, 1275VAC, 350VDC, 150uF	RV1	Wurth Electronics	820412711	Mouser	710-820412711	0.50	1
CMP-0136-00788	Low Quiescent Current LDO, 3-Pin SOT-23, Extended Temperature, Tape and Reel	U1	Murchip	MCPI700T-3002/TT	Arrow Electronics	MCPI700T-3002/TT	0.09	1
CMP-04889-0000	3V3QFN	U2	Texas Instruments	MPN302041TH8R	Mouser	595-MPN302041TH8R	2.64	1
6, 9610E-11	WR-F3H Fuseholder THT Brooks PCB Clip Cover	VF1	Wurth Electronics	6, 9610E-11	Mouser	710-69610E1003020	0.11	1
							81.29	61

Approved	Notes
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